

Total No. of Questions : 6]

SEAT No. :

P477

[Total No. of Pages : 3

APR - 18/BE/Insem. - 18

B.E. (Mechanical)

MECHANICAL SYSTEM DESIGN

(2012 Pattern) (Semester - II) (402048)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

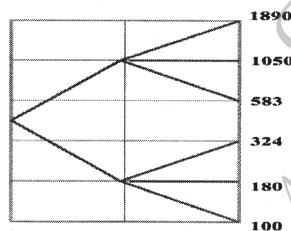
UNIT - I

- Q1)** a) What is structural formula? Write down any three structural formulae for a 12 speed gear box. **[4]**
- b) Draw the structural and speed diagram for a gear box having operating speed range 30 rpm to 700 rpm. Use R5 series, with standard spindle speeds. The gear box is connected to a motor driven by a pair of pulleys. Assume the motor speed to be 1440 rpm. **[6]**

OR

- Q2)** Read the structure diagram given below and answer the following questions **[10]**

- a) What is G.P. ratio & range ratio of this gearbox?
- b) Write structure formula for this gearbox?
- c) What is the speed of input shaft of the gearbox?
- d) Draw schematic layout diagram of the gearbox and calculate number of teeth on each gear by assuming 20 teeth on smallest gear of each stage.



P.T.O.

UNIT - II

- Q3)** a) Two populations X and Y are added together. Derive the expressions to find mean and standard deviations for the resultant population. [6]
- b) Explain design and natural tolerances. [4]

OR

- Q4)** The recommended class of fit for the journal and bearing of a hydrodynamic bearing is 50 H7e8. Diameters of journal and bearing are normally distributed. Maximum and minimum clearances are limited to 0.08 and 0.05 respectively. Determine number of rejected assemblies from a batch of 500 assemblies. Tolerances in microns are as follows : [10]

Diameter (mm)	H7		e8	
	Upper limit	Lower limit	Upper limit	Lower limit
40-60	+21	0	-40	-73

Area under normal distribution curve from 0 to Z is as follows :

Z	1	1.2	1.4	1.6	1.8	2	2.2	2.4	2.6	2.8
Area	0.341	0.384	0.419	0.445	0.464	0.477	0.486	0.491	0.495	0.497

UNIT - III

- Q5)** a) What are the types of load that can be handled by a material handling system? Explain the concept of containerization. [4]
- b) A three idler trough belt, horizontal conveyor is to be used for transporting 300 ton/hr of coal having weight density 9000 N/m³. The surcharge factor 'c' for three idler trough belt is 0.1. If belt speed is 120 m/min. Select the standard belt width for conveyor belt.

Available standard belt width is : 400, 450, 500, 650, 750, 800, 900, 1000, 1200, 1400, 1600, 1800, 2000 mm. [6]

OR

Q6) a) What do you understand by belt take-up system? State the different type of take-up systems used in belt conveyors and explain with neat sketch any one type take-up. [6]

b) Following data refers to a flat belt conveyor for transporting crushed rock. [4]

Mass Density = 2 tons/m³.

Belt speed = 1.75 m/s.

Belt width = 0.8 m

Flowability factor for surcharge angle 25° = 2.35×10^{-4}

Determine the capacity of conveyor in ton/hr.

